

Chapter 1: Introduction

Algorithms for Optimization

2nd Edition (2026)

Kochenderfer & Wheeler

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Chapter Overview

- 1 History of Optimization
- 2 The Optimization Process
- 3 Mathematical Formulation
- 4 Applications
- 5 Minima: Local vs. Global
- 6 Optimality Conditions
- 7 Overview of the Book
- 8 Summary & Exercises

Outline

- 1 History of Optimization
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A Brief History of Optimization I

Ancient Roots

- **Pythagoras / Euclid** (~300 BC): geometric extremum problems.
- **Archimedes** (~250 BC): maximizing the area of a polygon inscribed in a circle — a forerunner of *calculus of variations*.
- **Fibonacci** (125–1174 BC): *The Book of the Abacus* popularized the idea that mathematics could solve the world's problems.

A Brief History of Optimization II

Calculus Era (17th–18th Century)

- **Fermat** (1643): introduced the principle that *derivatives vanish at extrema* — first systematic derivative-based method.
- **Newton / Leibniz** (1660s–1680s): differential calculus provided tools to *identify stationary points* of functions.
- **Euler / Lagrange** (1740–1770s): *calculus of variations* — optimizing over functions rather than scalars; Lagrange multipliers for constrained optimization.

A Brief History of Optimization III

Modern Era

- **Cauchy** (1847): steepest-descent method — first numerical optimization algorithm.
- **Kantorovich** (1939): formulated *linear programming* for production planning; awarded Nobel Prize in Economics (1975).
- **Dantzig** (1947): *Simplex algorithm* for linear programming.
- **Kuhn & Tucker** (1951): KKT conditions for constrained optimization (now standard).
- **Computer age** (1950s–): large-scale optimization via iterative numerical algorithms.
- **Deep learning** (2012–): Watson/ChatGPT trained by optimizing networks with billions of parameters using stochastic gradient descent.

Outline

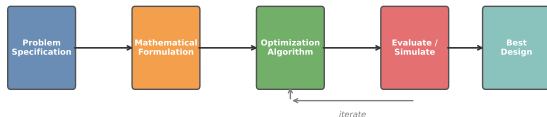
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The Optimization Process

Core Idea

Optimization is the process of **improving a system design** by adjusting its parameters to minimize (or maximize) a performance metric, subject to constraints.

The Optimization Process



Engineering Perspective

- 1 Specify the **design problem**.
- 2 Choose a **mathematical formulation**.
- 3 Select an **optimization algorithm**.
- 4 **Evaluate** the design (simulation / experiment).
- 5 **Iterate** until convergence.

Design Examples

- Aircraft geometry (airfoil parameters)
- Bicycle frame structure
- Autonomous vehicle controller
- Medical image classifier (neural network)

Optimization Algorithm Selection

Key Design Considerations

- **Dimensionality:** number of design variables n .
- **Gradient availability:** analytic vs. numerical vs. none.
- **Constraint structure:** unconstrained, bound, general.
- **Budget:** function evaluations may be expensive.
- **Noise:** deterministic vs. stochastic objective.
- **Multi-modality:** many local minima?

Method Families

Family	Example
First-order	Gradient descent
Second-order	Newton's method
Stochastic	SGD, Adam
Population	Genetic algorithms
Surrogate	Bayesian optim.
Direct	Nelder-Mead

No Free Lunch Theorem (informal)

No algorithm outperforms all others on every problem.
Algorithm choice must match problem structure.

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Mathematical Formulation

Standard Form of an Optimization Problem

$$\min_{\mathbf{x} \in \mathcal{X}} f(\mathbf{x}) \quad \text{subject to} \quad g_i(\mathbf{x}) \leq 0, \quad i = 1, \dots, m, \quad h_j(\mathbf{x}) = 0, \quad j = 1, \dots, p$$

- $\mathbf{x} = (x_1, x_2, \dots, x_n)^\top \in \mathbb{R}^n$
- $f : \mathbb{R}^n \rightarrow \mathbb{R}$
- $g_i(\mathbf{x}) \leq 0$
- $h_j(\mathbf{x}) = 0$
- $\mathcal{X}$

design point (decision variables)

objective function (minimize)

inequality constraints

equality constraints

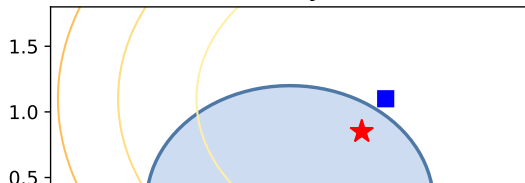
feasible set

Feasible Set

The **feasible set** $\mathcal{X}$ is the set of all design points satisfying every constraint:

$$\mathcal{X} = \{\mathbf{x} \mid g_i(\mathbf{x}) \leq 0, \quad h_j(\mathbf{x}) = 0, \quad \forall i, j\}$$

Feasible Set and Objective Contours



Feasible Set — Numerical Example

Example: 2D Constrained Problem

$$\min_{x_1, x_2} (x_1 - 2)^2 + (x_2 - 1)^2$$

subject to:

$$g_1 : x_1 + x_2 \leq 2, \quad g_2 : x_1 \geq 0, \quad g_3 : x_2 \geq 0$$

The **unconstrained minimum** is $(2, 1)$ — *infeasible* (violates g_1).

The **constrained minimum** is at the boundary of $\mathcal{X}$, found by solving the KKT conditions.

Plugging in boundary point $(1, 1)$:

$$f(1, 1) = (1 - 2)^2 + (1 - 1)^2 = 1$$

Design-Point Classification

Point	Status
$(0, 0)$	Feasible (corner)
$(1, 1)$	Feasible (on bound.)
$(2, 1)$	Infeasible
$(0.5, 0.5)$	Feasible (interior)

Modeling note

Constraints can be modeled as:

- **Inequality:** $g(\mathbf{x}) \leq 0$
- **Equality:** $h(\mathbf{x}) = 0$
- **Bound:** $x_i \in [l_i, u_i]$

Mathematical Formulation — Python Example

Listing 1: Constrained optimization with SciPy

```
1 import numpy as np
2 from scipy.optimize import minimize
3
4 # Objective:  $f(x) = (x_1-2)^2 + (x_2-1)^2$ 
5 def f(x):
6     return (x[0]-2)**2 + (x[1]-1)**2
7
8 # Constraints:  $x_1+x_2 \leq 2$ ,  $x_1 \geq 0$ ,  $x_2 \geq 0$ 
9 constraints = [
10     {'type': 'ineq', 'fun': lambda x: 2 - x[0] - x
11         [1]},
12 ]
13
14 bounds = [(0, None), (0, None)]
15
16 x0 = np.array([0.5, 0.5]) # initial guess
17 result = minimize(f, x0,
18                 method='SLSQP',
19                 bounds=bounds,
20                 constraints=constraints)
21
22 print("Optimal x:", result.x)
23 print("f(x*)      :", result.fun)
24 # Optimal x: [1. 1.]
25 # f(x*)      : 1.0
```

Key SciPy API

`scipy.optimize.minimize:`

- `method='SLSQP'`: Sequential Least Squares Programming (handles constraints)
- `bounds`: box constraints per variable
- `constraints`: list of dicts with 'type' ('eq'/'ineq') and 'fun'

Output

$x^* = [1.0, 1.0]$

$f(x^*) = 1.0$

Constraint g_1 : $1 + 1 = 2 \leq 2$ ✓

Active constraint!

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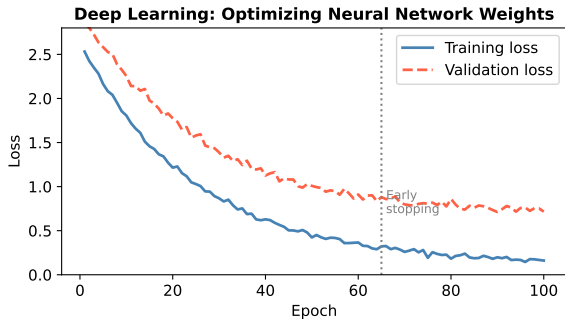
Applications of Optimization I

1.4.1 Aircraft Design

- **Objective:** minimize aerodynamic drag.
- **Design variables:** airfoil chord, thickness, sweep angle.
- **Constraints:** lift $\geq$ required, structural integrity.
- Evaluation via CFD simulation or wind-tunnel experiments.

1.4.2 Deep Learning

- **Objective:** cross-entropy loss over training data.
- **Variables:** billions of neural network weights.
- **Algorithm:** Stochastic Gradient Descent (SGD) / Adam.



Training and validation loss curves during neural network optimization.

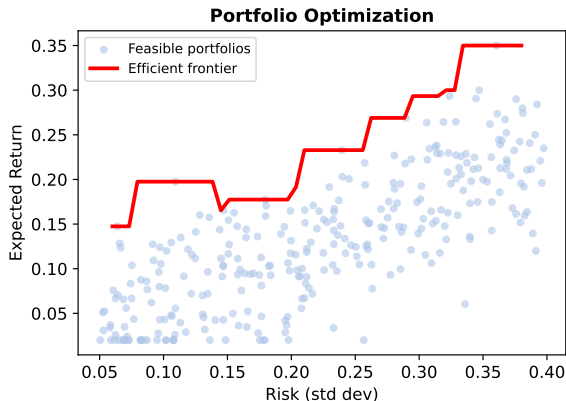
Applications of Optimization II

1.4.3 Financial Portfolio

- **Markowitz (1952)**: mean-variance portfolio optimization.
- **Objective**: maximize expected return for a given risk level.
- **Variables**: fraction of wealth in each asset, $\mathbf{x} \in [0, 1]^n$.
- **Constraint**: $\sum_i x_i = 1$ (budget), $x_i \geq 0$.
- **Efficient frontier**: set of Pareto-optimal portfolios.

Minimum-Variance Portfolio (Python)

Use `scipy.optimize.minimize` with equality constraint $\mathbf{1}^\top \mathbf{x} = 1$ and $\mathbf{x} \geq 0$.



Applications of Optimization III

1.4.4 Medical Radiation Therapy

- **Goal:** deliver prescribed radiation dose to tumor.
- **Constraint:** minimize damage to healthy tissue.
- **Formulation:** optimize beam intensities and angles.
- Objective involves a risk measure (CVaR) over dose distribution.

1.4.5 Job Scheduling

- **Goal:** assign n jobs to m machines to minimize makespan.
- **Combinatorial problem:** $n!$ possible orderings.

Cross-Domain Summary

Domain	Minimize	Variables
Aeronautics	Drag	Geometry
ML / DL	Loss	Weights
Finance	Risk	Allocation
Medicine	Toxicity	Beam angles
Manufacturing	Makespan	Schedule

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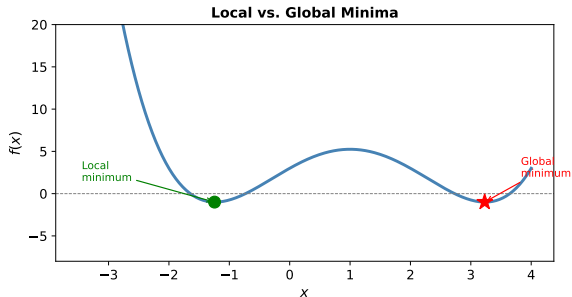
Types of Minima

Global Minimum

A point $\mathbf{x}^*$ is a **global minimum** if

$$f(\mathbf{x}^*) \leq f(\mathbf{x}) \quad \forall \mathbf{x} \in \mathcal{X}$$

- *Unique global minimum*: strict inequality $<$.
- Finding it is generally NP-hard for non-convex f .



Local Minimum

A point $\mathbf{x}^*$ is a **local minimum** if there exists $\delta > 0$ such that

$$f(\mathbf{x}^*) \leq f(\mathbf{x}) \quad \forall \mathbf{x} \in \mathcal{X}, \quad \|\mathbf{x} - \mathbf{x}^*\| < \delta$$

Numerical Example

$f(x) = \frac{x^3}{3} - x$ has a local minimum at $x = 1$ (not global — unbounded below).

Strong vs. Weak Minima

Classification

Type	Condition	Note
Strong local min	$f(\mathbf{x}^*) < f(\mathbf{x}), \mathbf{x} \neq \mathbf{x}^*$	Unique locally
Weak local min	$f(\mathbf{x}^*) \leq f(\mathbf{x})$	Plateau possible
Global min	$f(\mathbf{x}^*) \leq f(\mathbf{x}), \forall \mathbf{x} \in \mathcal{X}$	Hard to find

Practical Implication

Most optimization algorithms **converge to local minima**. Strategies to escape:

- Multi-start (random restarts)
- Simulated annealing / population methods
- Convex relaxation

Exercise 1.1 (Book)

Q: Give a function with a local min that is not global.

A: $f(x) = x^3/3 - x$

Local min at $x = 1$: $f(1) = -2/3$.

But $f(x) \rightarrow -\infty$ as $x \rightarrow -\infty$,
so $x = 1$ is not the global minimum.

Exercise 1.2 (Book)

Q: Minimizers of $f(x) = x^3 - x$?

A: No minimum — unbounded below.

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1.6 Optimality Conditions — Univariate I

Setting

Consider **unconstrained** minimization of $f : \mathbb{R} \rightarrow \mathbb{R}$. Necessary conditions for x^* to be a local minimum:

First-Order Necessary Condition (FONC)

$$f'(x^*) = 0$$

A point satisfying FONC is called a **stationary point**. Not all stationary points are minima (could be max or inflection).

Second-Order Necessary Condition (SONC)

$$f''(x^*) \geq 0$$

Together with FONC, these are *necessary* — all local minima must satisfy them.

Sufficient Condition for Strong Local Min

$$f'(x^*) = 0 \quad \text{and} \quad f''(x^*) > 0$$

$f''(x^*) > 0$ ensures the stationary point is a bowl (locally convex).

Key Insight via Taylor Expansion

From $f(x^* + h) = f(x^*) + hf'(x^*) + O(h^2)$:

$$f(x^* + h) \geq f(x^*) \implies hf'(x^*) \geq 0$$

$$f(x^* - h) \geq f(x^*) \implies hf'(x^*) \leq 0$$

1.6 Optimality Conditions — Univariate II

Taylor Expansion (Eq. 1.4–1.8)

$$f(x^* + h) = f(x^*) + hf'(x^*) + O(h^2) \quad (1.4)$$

$$f(x^* - h) = f(x^*) - hf'(x^*) + O(h^2) \quad (1.5)$$

$$f(x^* + h) \geq f(x^*) \implies hf'(x^*) \geq 0 \quad (1.6)$$

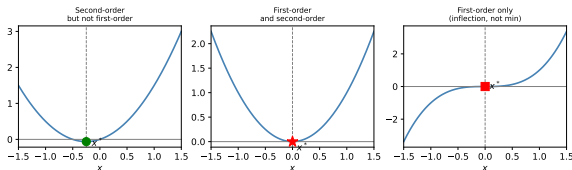
$$f(x^* - h) \geq f(x^*) \implies hf'(x^*) \leq 0 \quad (1.7)$$

$$\implies f'(x^*) = 0 \quad (1.8)$$

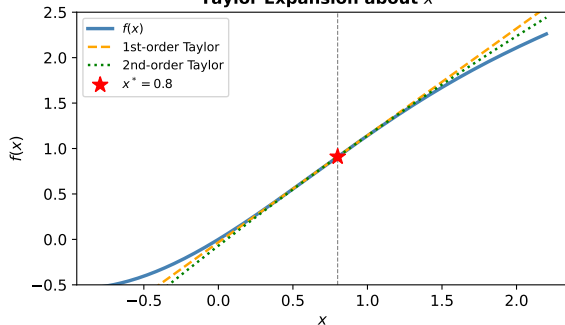
Second-Order (Eq. 1.9–1.10)

$$f(x^* + h) = f(x^*) + \underbrace{hf'(x^*)}_{=0} + \frac{h^2}{2}f''(x^*) + O(h^3)$$

Necessary Conditions for Local Minima (Univariate)



Taylor Expansion about x^*



1.6 Optimality Conditions — Multivariate I

Extension to $f : \mathbb{R}^n \rightarrow \mathbb{R}$

For unconstrained $\min_{\mathbf{x}} f(\mathbf{x})$, the necessary conditions generalize:

First-Order Necessary Condition

$$\nabla f(\mathbf{x}^*) = \mathbf{0}$$

The **gradient** at a local minimum is zero.

$$\nabla f(\mathbf{x}) = \left[\frac{\partial f}{\partial x_1}, \dots, \frac{\partial f}{\partial x_n} \right]^\top$$

Second-Order Necessary Condition

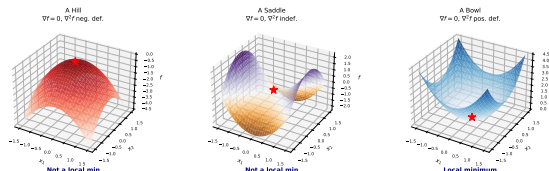
$$\nabla^2 f(\mathbf{x}^*) \succeq \mathbf{0} \quad (\text{positive semidefinite})$$

The **Hessian** matrix:

Three Possible Regions at $\nabla f = 0$

- 1 **Hill:** $\nabla^2 f \prec \mathbf{0}$ (neg. def.) — local maximum
- 2 **Saddle:** $\nabla^2 f$ indefinite — not a min/max
- 3 **Bowl:** $\nabla^2 f \succ \mathbf{0}$ (pos. def.) — local minimum

Three Local Regions Where $\nabla f = 0$ (Figure 1.15)



1.6 Optimality Conditions — Multivariate II

Example 1.2 (Book): Rosenbrock Function

$$f(\mathbf{x}) = (1 - x_1)^2 + 5(x_2 - x_1^2)^2$$

Check whether $[1, 1]$ satisfies FONC and SONC.

Gradient:

$$\nabla f(\mathbf{x}) = \begin{bmatrix} 2(10x_1^3 - 10x_1x_2 + x_1 - 1) \\ 10(x_2 - x_1^2) \end{bmatrix}$$

At $[1, 1]$: $\nabla f(1, 1) = \mathbf{0} \checkmark$

Hessian:

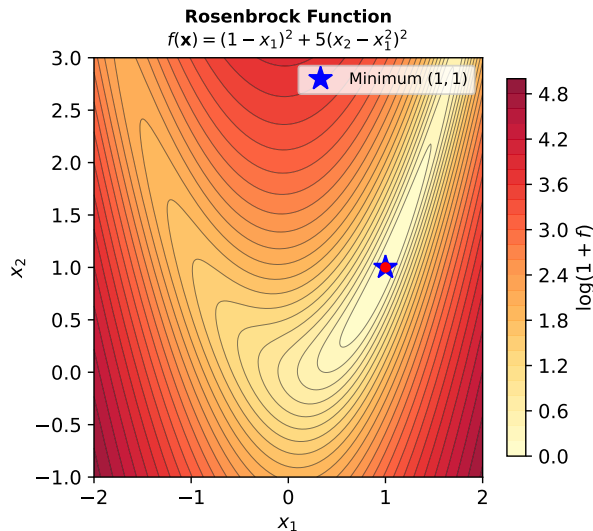
$$\nabla^2 f(\mathbf{x}) = \begin{bmatrix} -20(x_2 - x_1^2) + 40x_1^2 + 2 & -20x_1 \\ -20x_1 & 10 \end{bmatrix}$$

At $[1, 1]$:

$$\nabla^2 f(1, 1) = \begin{bmatrix} 42 & -20 \\ -20 & 10 \end{bmatrix}$$

Eigenvalues > 0 (positive definite) $\checkmark$
 $\det = 42 \cdot 10 - (-20)^2 = 20 > 0 \checkmark$

1.6 Optimality Conditions — Multivariate III



Rosenbrock Contour

- Narrow curved valley — *banana-shaped*
- Minimum at $(1, 1)$: $f(1, 1) = 0$
- Classic benchmark for optimization algorithms
- Gradient can be misleading due to narrow valley

Example 1.1: Visualization

For $f(x_1, x_2) = x_1^2 - x_2^2$:

- Saddle at origin: $\nabla f = \mathbf{0}$
- $\nabla^2 f = \text{diag}(2, -2)$ — indefinite
- Not a minimum!

Optimality Conditions — Python Verification

Listing 2: Checking optimality conditions (Python)

```
1 import numpy as np
2
3 def rosenbrock(x):
4     return (1 - x[0])**2 + 5*(x[1] - x[0]**2)**2
5
6 def grad_rosenbrock(x):
7     x1, x2 = x
8     df_dx1 = 2*(10*x1**3 - 10*x1*x2 + x1 - 1)
9     df_dx2 = 10*(x2 - x1**2)
10    return np.array([df_dx1, df_dx2])
11
12 def hess_rosenbrock(x):
13     x1, x2 = x
14     h11 = -20*(x2 - x1**2) + 40*x1**2 + 2
15     h12 = -20*x1
16     h22 = 10
17    return np.array([[h11, h12], [h12, h22]])
18
19 xstar = np.array([1.0, 1.0])
20 g = grad_rosenbrock(xstar)
21 H = hess_rosenbrock(xstar)
22 eigs = np.linalg.eigvalsh(H)
23
24 print("Gradient:", g)           # [0. 0.]
25 print("Hessian:\n" + H)        # [[42. -20.]
```

Output

Gradient: [0. 0.]

Hessian:

[[42. -20.]
[-20. 10.]]

Eigenvalues: [2.0, 50.0]
(all positive)

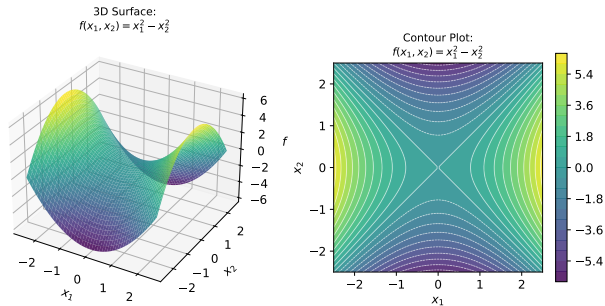
FONC satisfied: True

SONC satisfied: True

Conclusion

[1, 1] is a **strong local minimum** of the Rosenbrock function (in fact, the global minimum with $f^* = 0$).

Surface and Contour Visualization



Listing 3: 3D surface + contour (Python)

```
1 import numpy as np
2 import matplotlib.pyplot as plt
3 from mpl_toolkits.mplot3d import Axes3D
4
5 x1 = np.linspace(-2.5, 2.5, 100)
6 x2 = np.linspace(-2.5, 2.5, 100)
7 X1, X2 = np.meshgrid(x1, x2)
8 Z = X1**2 - X2**2 # saddle
9
10 fig = plt.figure(figsize=(10, 4))
11 # --- 3D ---
12 ax3d = fig.add_subplot(121, projection='3d')
13 ax3d.plot_surface(X1, X2, Z, cmap='viridis')
14 ax3d.set_xlabel('$x_1$'); ax3d.set_ylabel('$x_2$')
15 # --- 2D contour ---
16 ax2d = fig.add_subplot(122)
17 ax2d.contourf(X1, X2, Z, 20, cmap='viridis')
18 ax2d.set_xlabel('$x_1$'); ax2d.set_ylabel('$x_2$')
19 plt.tight_layout()
```

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1.7 Overview of the Book I

Part I: Derivatives and Bracketing

- Ch. 2 Derivatives and Gradients
- Ch. 3 Bracketing (1D search)
- Ch. 4 Local Descent

Part II: First & Second Order

- Ch. 5 First-Order Methods (gradient descent, subgradient)
- Ch. 6 Second-Order Methods (Newton, quasi-Newton)
- Ch. 7 Conjugate Direction Methods
- Ch. 8 Stochastic Methods (SGD, Adam)
- Ch. 9 Noisy Objectives

Part III: Constrained Optimization

- Ch. 10 Constraints (KKT, penalty methods)
- Ch. 11 Linear Programming (Simplex, interior point)
- Ch. 12 Network Flow
- Ch. 13 Sampling Plans
- Ch. 14 Surrogate Models

Part IV: Advanced Topics

- Ch. 17 Surrogate Optimization
- Ch. 20 Uncertainty in Optimization
- Ch. 21 Multi-Objective Optimization
- Ch. 23 Expression Optimization
- Ch. 24 Multidimensional Design

1.7 Overview of the Book II

Derivative-Based Methods (Chs. 5–9)

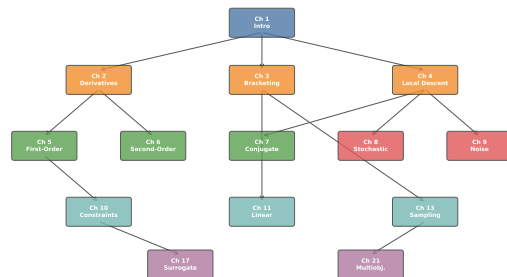
Ch. 5 covers *first-order* gradient methods. Ch. 6 builds *second-order* methods using the Hessian. Ch. 7 introduces conjugate directions to avoid redundant search. Ch. 8 covers *stochastic* gradient for noisy/large-scale settings.

Stochastic & Surrogate Methods

When function evaluations are expensive, we use:

- **Surrogate models** to approximate f cheaply.
- **Bayesian optimization**: model uncertainty about f , balance exploration vs. exploitation.

Chapter Dependency Overview (simplified)



Chapter dependency graph (simplified). Reading order follows arrows.

1.7 Overview of the Book III

Constrained Optimization (Ch. 10–)

Chapter 10 introduces constrained optimization and KKT conditions (which generalize the unconstrained optimality conditions of Ch. 1). Linear programming (Ch. 11) is the special case where both f and constraints are linear.

Uncertainty and Multiple Objectives

- Ch. 20: probabilistic & robust optimization.
- Ch. 21: multi-objective (Pareto front, scalarization).
- Ch. 22: discrete / combinatorial optimization.

Algorithm Families at a Glance

Family	Info Used	Key Ch.
Gradient	∇f	5, 6
Newton	$\nabla^2 f$	6
Conjugate	History	7
Stochastic	Noisy ∇f	8, 9
Evolutionary	f only	14
Bayesian	Posterior	17

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1.8 Summary

Key Takeaways

- 1 Optimization = finding the best design subject to constraints.
- 2 Standard form:

$$\min_{\mathbf{x} \in \mathcal{X}} f(\mathbf{x})$$

- 3 A **local minimum** satisfies $f(\mathbf{x}^*) \leq f(\mathbf{x})$ in a neighbourhood.
- 4 **Minima require:** $\nabla f = \mathbf{0}$ (FONC) and $\nabla^2 f \succeq 0$ (SONC).
- 5 Zero gradient does *not* imply a minimum — check the Hessian!
- 6 Most algorithms converge to *local* minima; escaping requires global strategies.

Core Formulas to Remember

$$\text{FONC: } f'(x^*) = 0 \quad [\text{or } \nabla f(\mathbf{x}^*) = \mathbf{0}]$$

$$\text{SONC: } f''(x^*) \geq 0 \quad [\text{or } \nabla^2 f(\mathbf{x}^*) \succeq 0]$$

$$\text{Strong local min: } f'(x^*) = 0, f''(x^*) > 0$$

Applications span

Engineering design · Machine learning · Finance ·
Medicine · Manufacturing · Robotics ...

1.9 Exercises (with Solutions) I

Exercise 1.1

Q: Give an example of a function with a local minimum that is not a global minimum.

A: $f(x) = x^3/3 - x$. Local min at $x = 1$ ($f''(1) = 2 > 0$), but $f(x) \rightarrow -\infty$ as $x \rightarrow -\infty$.

Exercise 1.2

Q: What is the minimum of $f(x) = x^3 - x$?

A: No minimum — the function is unbounded below.

Exercise 1.3

Q: Give a function bounded below but with no global optimum on $\mathcal{X} = [0, 1]$.

A: $f(x) = \begin{cases} |x| & x \neq 0 \\ 1 & x = 0 \end{cases}$. The infimum is 0 but it is not attained.

1.9 Exercises (with Solutions) II

Exercise 1.4

Q: What are the minima and minimizers of $f(x) = \sin(x)$?

A: Minimum value = -1 , at $x = \frac{3}{2}\pi + 2\pi i$ for all integers i . Infinitely many global minimizers.

Exercise 1.5

Q: Does the first-order condition $f'(x) = 0$ hold at the optimal solution of a constrained problem?

A: No. Counterexample: $\min f(x) = x$ subject to $x \geq 1$. Optimal at $x = 1$, but $f'(1) = 1 \neq 0$.
(Constrained conditions require KKT — Chapter 10.)

Exercise 1.6

Q: How many minima does $f(x, y) = x^2 + y$ subject to $x > y \geq 1$ have?

A: None. $g(x) = x^2$ has infimum at $x \rightarrow 1$, $h(y) = y$ increases for $y \geq 1$. But $x > y$ (strict), so we can only approach $x = y = 1$, never attain it.

1.9 Exercises (with Solutions) III

Exercise 1.7

Q: How many inflection points does $f(x) = x^3 - 10$ have?

A: One. $f''(x) = 6x = 0 \Rightarrow x = 0$. Sign of f'' changes at $x = 0$ (negative for $x < 0$, positive for $x > 0$), confirming exactly one inflection point.

Study Tips

- Always check both FONC *and* SONC when identifying minima.
- For constrained problems, FONC becomes the KKT conditions (Chapter 10).
- Use Python's `scipy.optimize.minimize` with `method='L-BFGS-B'` (unconstrained) or `'SLSQP'` (constrained) as your go-to numerical solver.

Chapter 1: Introduction

Algorithms for Optimization

Kochenderfer & Wheeler, 2nd Ed. (2026)

What we covered

- History of optimization
- The optimization process
- Mathematical formulation
- Applications (engineering, ML, finance)
- Local vs. global minima
- Optimality conditions (FONC, SONC)
- Overview of the book's chapters

Up Next: Chapter 2

Derivatives and Gradients

- Finite difference methods
- Automatic differentiation
- Complex-step differentiation
- Directional derivatives